The Interplay Between Efficient Representation Learning and Optimization Algorithms

It is not an overstatement to say that the success deep learning can be largely attributed to *representation learning* (Bengio et al., 2013). The precise notion and objective behind "representation" can vary between communities, but even from a purely function-approximation point of view, deep learning methods have demonstrated an astonishing ability to grasp predictive features for complex tasks with minimal *a priori*, human-specified structure. In fact, on the empirical frontier of deep learning, it is a recurring theme that the best practice is for the engineer to loosen their specification on the *model* and instead focus their efforts on cleaning and curating the data. This is best evidenced by modern large-language models (LLM) applied to natural language tasks, which share almost all their important attributes with the original Transformer architecture proposed in Vaswani (2017), such that the bulk of their impressive performance solely comes from the scale of the data and model size.

On the flipside, the incredible growth of deep learning left many theorists in the dust, as it seemed to defy every facet of established theoretical dogma. In particular, neural networks as a function class induce highly non-convex optimization landscapes and exhibit prohibitively large complexity for generalization measured by standard learning-theoretic quantities (e.g. VC dimension, Rademacher complexity). Yet, adequately trained neural networks seem to experience few issues reaching good optima, and enjoy excellent generalization that seems to violate the bias-variance tradeoff. Fortunately, in the ensuing years since the deep learning boom, theoreticians have steadily demystified pieces of the puzzle, including but not limited to: identifying large classes of problems where the non-convexity does not preclude convergence (e.g. benign overparameterization) (Bartlett et al., 2020; Lee et al., 2016), updating classical notions of bias-variance (Yang et al., 2020), and refining data-dependent complexity measures (Lotfi et al., 2022). These form solid foundations on which to theoretically analyze deep learning problems.

However, one of the remaining fundamental questions in the intersection of both theory and practice is the following: are standard black-box *first-order* optimization methods (notably Adam) truly well-suited for representation learning? From a practical point of view, it remains surprising that local search methods on the raw model parameter space are capable of finding rich structure (compositional, hierarchical etc.) at all. Therefore, a burgeoning research area has aimed to revisit optimization principles for deep learning, since any improvement in convergence and generalization tangibly amounts to millions of dollars in saved compute and energy. Encouragingly, the winner of a recent MLCommons Optimization competition is a pseudo-*second-order* method, beating out all hyperoptimized Adam variants. From a theoretical point of view, higher-order optimization methods are by definition better suited to following the natural geometry of the optimization landscape, with computational complexity and stochasticity being the major hurdles to make them practical for internet-scale problems. However, recent work (Zhang et al., 2024b) has demonstrated that not only are higher-order methods better-suited, but they may be *necessary* for detecting an optimal representation, even in simple *linear* settings. These two threads spanning empirical to theoretical suggest there is a unifying theme to be uncovered with

regard to higher-order information and representation learning. Toward following these threads, the tangible goals of this independent study are the following:

- Survey various areas of recent literature, including feature learning, multi-task learning (Du et al., 2020; Zhang et al., 2024a), and (two-timescale) optimization dynamics (Doan, 2022; Marion and Berthier, 2024).
- Construct a family of test problems to ablate the representation learning capabilities and computational burden of various proposed methods, starting from problem classes contained in Zhang et al. (2024b).
- Using Zhang et al. (2024b) as a starting point, where representation learning is cast as a linear regression, initialize theoretical study into aforementioned optimization methods to rigorously capture failure/success of representation learning.

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